

How Do First-Year Engineering Students' Emotions Change while Working on Programming Problems?

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Emotions are a complex multi-faceted phenomenon. To assess the complexity of emotions from different facets, multi-modal approaches are necessary. However, multi-modal approaches are rarely used for assessing emotions, especially in the context of computer programming. This study adopts a multi-modal approach to understand the changes in students' perception of emotions before and after working on programming problems. Understanding these changes in students' perceptions may enable educators to devise interventions that help students adjust their perceptions and regulate their emotions as per their skills. We conducted a one-on-one programming session and retrospective think-aloud interview with 17 students from an introductory programming course. During the programming session, students filled surveys and performed four programming tasks. While working on these tasks, students' eye gaze, video of face and screen, and electrodermal activity data were also collected using a non-invasive device. The data collection for this study was multi-modal, with a mix of both qualitative and quantitative data collection methods. Data analysis was primarily qualitative, with additional triangulation of qualitative and biometric data for select exemplars. The findings of this study suggest that students experience changes in emotions because of many reasons, for instance, they encountered repeated errors, they set high standards for their performance, or they could not manage time. For some students, negative emotions changed to positive emotions when they solved errors without any external help or achieved more than what they expected going into the task. Moreover, the triangulation of qualitative and biometric data of two participants provides a fine-grained analysis of their emotions and behaviors and confirmed the change in the perception of their emotions while performing the programming tasks.

CCS Concepts: • **Social and professional topics** → **Computing education**;

Additional Key Words and Phrases: Emotions, multi-modal data, and programming

ACM Reference Format:

Zahra Atiq and Rakhi Batra. 2024. How Do First-Year Engineering Students' Emotions Change while Working on Programming Problems?. *ACM Trans. Comput. Educ.* 24, 2, Article 24 (April 2024), 30 pages. <https://doi.org/10.1145/3643865>

This work is supported in part by the Dale and Suzi Gallagher Professorship in Engineering Education, Bilsland Dissertation Fellowship at Purdue University, and the National Science Foundation under the grant (NSF IIS 2104729). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

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ACM 1946-6226/2024/04-ART24

<https://doi.org/10.1145/3643865>

1 INTRODUCTION

Most of the undergraduate programs related to **Science Technology Engineering and Mathematics (STEM)** require novice students to take introductory programming courses. In these courses, students learn to write computer programs where they often face difficulties, because novice students normally have inaccurate mental models of programming concepts [28, 59]. Due to these difficulties, students feel negative emotions, which in turn impact their learning and performance [53, 74]. Many researchers have investigated the effect of emotions on students' motivation and learning [36, 37, 41, 58] but a fine-grained understanding of emotions in the context of computer programming is very limited.

Moreover, it is not well understood why students' emotions change as they work on programming problems. The important mediating factors could be students' self-concept of ability, outcome expectations, interest in programming, difficulty in learning problem-solving skills, and coping strategies [30, 50, 65]. An understanding of these reasons could help educators design interventions required to help novice students learn programming effectively. These interventions could be providing feedback after failure, developing coping and problem-solving strategies, appraisal-oriented regulation (restructuring expectations and beliefs), and acquiring the necessary programming skills (writing code and debugging) [50, 65]. Hence, it is important to understand the reasons why there is a change in students' perception of emotions before and after completing a programming task. In this study, our focus is to answer these research questions: (1) How do students' emotions change while working on programming problems? and (2) What are some reasons students provide for the change in their emotions while working on programming problems?

This work is part of a larger study, the purpose of which is to understand the emotions first-year engineering students experience while working on programming problems. The current study adopts a multi-modal approach to answer the research question, because multi-modal approaches help capture multi-layered information about the complexity of human emotions [1, 71]. The types of data we have collected include self-report (retrospective think-aloud interview, **Achievement Emotion Questionnaire for Programming (AEQp)**) and one physiological biomarker (**electrodermal activity (EDA)**). The data analysis methods are primarily qualitative followed by triangulation with other modes of data for two participants.

The rest of the article is organized as follows: Section 2 describes the conceptual framework used to guide different stages of the research process. Section 3 presents the literature review on change in emotions in the context of education, specifically during programming, Section 4 explains the research design, data collection and analysis methods, and experimental setup. The findings of the study are explained in Section 5 and Section 6 presents the discussion of findings. Limitations and future directions and implications for classroom practice are discussed in Sections 7 and 8, respectively. Finally, Section 8 presents a conclusion, highlighting the important findings, implications, and limitations of the study.

2 CONCEPTUAL FRAMEWORK

2.1 Contextualizing Emotions in Education

Emotions are psychological processes, composed of multiple components (motivational, physiological, cognitive, affective, and expressive) [62]. Each component contributes uniquely while a person experience certain emotions. To understand these components, consider a student who feels shame after failing an exam. Shame may comprise feeling ashamed (affective), gaze aversion (physiological), avoidance behavior (motivation), low self-worth (cognitive), and changed facial expressions (expressive). Emotions play a vital role in the academic achievement, motivation, self-efficacy, and psychological well-being of students. Different students may feel different emotions

while facing the same situation because of their cultural differences and personal histories [74]. Therefore, it is important to understand emotions and how they affect students in different educational contexts [58].

2.2 Control-Value Theory of Achievement Emotions

Control-Value Theory of Achievement Emotions (CVT) offers an integrated framework for understanding emotions in academic settings, hence, CVT is suitable to answer the research question of this study. CVT posits that students' emotions influence their performance and learning through motivational and cognitive mechanisms [58]. This theory is based on the idea that students feel positive emotions when they are engaged in a learning activity that they find valuable or feel in control of that activity. Similarly, students feel negative emotions when they engage in a learning activity that they do not find valuable, or do not feel in control of, while performing that activity. A combination of different control and value appraisals leads to experiencing different emotions. CVT also sets forth the idea that emotions, their triggers, and outcomes are linked together by reciprocal causation, resulting in a feedback loop. The feedback loop subsequently implies that students' emotions could be regulated by targeting any one of the three elements of the feedback loop [58].

An important component of CVT is self-concept, which is defined as the cognitive appraisal people hold about their own qualities, perceptions, and expectations [29]. Self-concept is conceptualized as a multidimensional, context-dependent, and hierarchical phenomenon [64]. CVT also describes the self-concept of ability as the judgment of one's ability in a certain domain. Students' self-concept of ability does not describe the causal relationship between ability judgment and emotions experienced. However, the self-concept of ability is the foundation of perceived control in the achievement domain [58].

3 LITERATURE REVIEW

3.1 Change in Emotions in Education

In academics, students' emotions have been shown to relate to important processes and outcome variables like academic achievement, learning strategies, long-term learning, and career choices [26, 52]. Emotions are momentary, because they occur as a response to a situation at a specific time. For instance, most students feel anxious before an exam and relieved after the exam [58]. Similarly, Spangler et al. investigated how students' emotional reactions and their coping strategies change before and after an exam [65]. The authors found that hopelessness and anxiety were prominent before the exam, but these emotions dissipated as the exam ended [65]. Moreover, students with high emotion-oriented coping skills experienced fewer negative emotions and vice versa [65]. These changes in emotions could be due to the situational characteristics of different phases of the exam. However, students may also experience emotions based on their general perception of a situation, beliefs, and experiences [50]. For instance, Bieg et al. conducted a study to investigate whether there is a difference between students' general perception of their emotions and the emotions they actually feel in the mathematics course. Their findings suggest that there is a difference between how students think they feel and what they actually feel, because their perception is based on their beliefs and generic experiences, while actual emotions are triggered by the situation at hand [9].

Our synthesis of literature suggests that students experience changes in their emotions as they go through different academic situations. This argument forms the basis to understand the change in emotions in a specific domain such as computer programming.

3.2 Change in Emotions in the Context of Learning Programming

In recent years, some research studies have emerged that focus on the changes in students' emotions in the context of computer programming [11, 36, 37, 41, 42, 58]. Kinnunen and Simon developed a framework that helps to understand the students' emotions and how these emotions change during different stages of programming in a CS1 course [36, 37]. For instance, during the "Getting Started" stage, students did not know how to solve the problem, and they felt puzzled or confused, whereas, in the "Succeeding" stage, students felt relief or accomplished if they solved the problem successfully. The change in students' emotions is due to their perception of the ability to perform the task and the task outcome. Their perception changes as the students acquire subject knowledge and develop self-regulation strategies. Lishinski et al. in their recent study investigated how the changes in students' affective experiences and self-efficacy develop their long-term interest in computing [40]. The authors found that students' affective experiences change over the semester and impact their interest and performance in the course. Angel and Daniel conducted a study to understand the emotional impact of using educational software (OptimEx) in an Algorithm course offered in-person and online [66]. The authors used the PANAS questionnaire to collect students' emotional responses before and after solving an algorithm task using OptimEx. Their results suggest that students' negative emotional levels decreased after the task and students who attended in person experienced an increase in their positive emotions after the task.

Some researchers have also used fine-grained data on student interactions to understand how students' emotions change while working on programming problems. Mercedes et al. collected data from freshman computer science students during lab sessions, over a period of nine weeks in a CS1 course [60, 61]. The authors found that affective states of confusion, neutrality, and flow changed over time and frustration varied significantly [61]. Since the studies conducted by Mercedes et al. consider observational data and students' activity logs, they miss out on the students' self-reported emotional experiences. Hence, it would be useful to ask students about their perceptions of emotional experiences to enhance the significance of the findings. Using students' reflections, Kellam et al. investigated the change in students' emotions after they participated in different activities and projects over a semester. They performed a narrative analysis of student reflections and found that students experience anxiety and frustration at the start of the semester but as the students' progress and learn self-regulation strategies, they start feeling positive emotions [33]. This study only focused on student reflections, which covered one perspective of emotions. It is possible that students may fail to recall their individual emotional experiences over the course of the semester and report the most recent or generic emotions. In this context, D'Mello et al. studied novice students' emotions during two programming phases (namely scaffolding phase and fade-out) by analyzing their self-reported emotional judgments. They identified that most students' emotions transition from engagement to confusion, confusion to frustration, and frustration to boredom while programming [12, 13].

From these studies, it is evident that students' emotions change during an exam or programming task. The change could be due to students' emotional regulation strategies, coping strategies for emotional balance, lack of ability and control over outcomes, and restricted resources for solving the problems [58, 65]. To avoid the negative consequences of these emotional changes on students' learning and performance, it is important to understand the change in emotions and the corresponding reasons.

3.3 Using Biometrics to Understand Students' Emotions

With the development of modern neuroscience, researchers have begun to use physiological measures such as EDA, heart rate, and brain activity to understand human emotions [24, 38, 73]. For

instance, EDA measures skin conductance by counting the number of active sweat glands [14]. The change in the number of sweat glands is the result of activation of the **sympathetic nervous system (SNS)**, which is a physiological response to stress conditions (also called fight or flight response) [14]. The physiological response through SNS activation represents arousal response in the form of EDA [67]. Hence, the EDA levels denote near real-time arousals that may be due to emotional reactions or performing cognitively challenging tasks [14].

Villanueva and colleagues conducted a few studies that employ electrodermal activity along with other data collection methods to study the change in emotions of students [37–39]. Results from these studies suggest that there is a weak positive correlation between EDA and difficulty of the exam and between temperature and exam. These findings indicate a possible connection between cognition and emotion via physiological measures.

Girardi et al. have used electroencephalography, heart rate, and blood volume pulse to identify emotions experienced by the developers and the events that trigger those emotions [25]. They found that although developers experience a wide range of emotions during programming tasks, emotions related to negative valence and high arousal were most common. These emotions were triggered by unexpected code behavior, missing documentation, time pressure, and self-perceived low productivity. Fritz and Muller conducted multiple studies to analyze the perceived progress of programmers and task difficulty by using the biometric data [23, 46, 47]. They found that programmers experience many emotions that affect their perceived progress and productivity. Furthermore, they suggested that identifying the emotions of programmers may help to improve their productivity by providing suggestions and resources when they are found stuck and make no progress. Recently, Gorson et al. used electrodermal activity to study the undergraduate programming students' emotional experiences [27]. They collected students' EDA data while they were working on a programming task. After the task, the authors used the collected EDA data to help the participants in recalling their programming emotions during a retrospective interview. Using the interview data and patterns in EDA, authors identified programming events that triggered emotions like anxiety. The findings from this study support the use of EDA data to study emotional arousal during programming.

3.4 Summary of Current Research

In literature, researchers have investigated students' emotions and how these emotions change by employing different data collection methods such as self-report questionnaires and interviews, physiological measures, and programming activity logs. Most of these studies either use one source of data or are not in the context of computer programming. However, the work of Villanueva et al. has demonstrated a significant contribution to understanding students' emotions by using multi-modal data [34, 68]. The multimodal data provides a fine-grained and near real-time understanding of students' emotions and helps in increasing the validity of findings [2]. Hence, it is appropriate to use multimodal data to understand the change in students' emotions as they work on a programming task. In this study, we collected eye gaze, EDA, and retrospective think-aloud interviews (self-report) data. The data collection for this study is multi-modal. However, data analysis is primarily qualitative, with a follow-up multi-modal phase meant for triangulation purposes.

4 RESEARCH METHODOLOGY

This study is part of a larger project and a more detailed description of the research methodology can be found in our previous publication [3]. However, we provide a complete summary of the methodology in the following sections.

Table 1. Self-reported Demographic Information of the Study Participants

Pseudonym	Age	Gender	Race/Ethnicity	Pseudonym	Age	Gender	Race/Ethnicity
Andrew	19	Male	Black/African American	Jack	18	Male	Hispanic/Latinx
Anna	19	Female	White	John	18	Male	White
Becky	19	Female	White	Lilly	18	Female	White
Christina	18	Female	White	Mark	19	Male	White
Ella	19	Female	Asian	Martha	19	Female	White
Emily	18	Female	White	Rachel	19	Female	White
Erica	19	Female	White	Randy	18	Male	Hispanic/Latinx
George	19	Male	American Indian/Alaskan	Sarah	18	Female	Asian
Harris	19	Male	Asian				

4.1 Context of the Study

This study is conducted at a midwestern research-intensive university in the United States. Every year, around 2500 new undergraduate engineering students get enrolled in its **first-year engineering (FYE)** program. The students in FYE program take two fundamental STEM courses that contain topics from computer programming, data visualizations, engineering design, teamwork, and communications. The context of this study is one of the two STEM courses (ENGR 132). ENGR 132 is a two-credit course that focuses on introductory programming concepts such as logical operators, conditional statements, for and while loops, nested loops, mathematical calculations, and basic data visualization techniques in MATLAB. Moreover, ENGR 132 has multiple sections and each section of ENGR 132 has up to 120 students.

4.2 Selection Criteria and Participants

For this study, we recruited 17 participants using purposive criterion sampling [49]. During the Spring 2018 semester, these participants were recruited from seven sections with a total enrollment of 768 students. Purposive criterion sampling ensures that we capture rich and multiple perspectives about participants [39]. For this study, we collected data after Spring break. By that time, students had covered the concepts of simple and nested loops. We included novice students who were taking ENGR 132 for the first time. For this study, we specifically excluded students repeating ENGR 132, students taking another programming course in parallel, and international students. Participants’ demographic information is provided in Table 1. To ensure confidentiality, we anonymized participants by assigning them pseudonyms. Each participant was emailed a \$40 Amazon gift card as compensation for their participation.

4.3 Description of the Programming Task

Participants were given 30 minutes to complete four problems on the programming task. These problems were selected by discussing the difficulty level with an instructor who had taught and designed ENGR 132 in MATLAB for many years. Students had the option to work on the problems in any order and could either use built-in MATLAB functions or write code for a problem. A brief description of programming problems is given in Table 2.

4.4 Research Design

For this study, we used a multi-modal approach for data collection and data analysis [2]. Multi-modal research designs are versatile, because other methodologies and methods can be embedded

Table 2. Brief Description of the Programming Tasks

Problem	Description
1	An engineering student and his family are planning activities for their summer vacation to the Indiana Dunes & Lake Michigan. The student needs a MATLAB function to help them decide what to do. Here are the criteria: 1. If the temperature is greater than or equal to 90 °F, they will swim. 2. If the temperature is greater than or equal to 80 °F and less than 90 °F, they will go boating. 3. If the temperature is less than 80 °F, they will go fishing. Write the MATLAB function to determine which activity the family will do based on the outside temperature, and then display the decision to the MATLAB command window. Include appropriate comments with your code. Test your function using the temperatures 70 °F, 85 °F, and 90 °F.
2	Given the vector $x = [1 \ 8 \ 3 \ 9 \ 0 \ 1]$, write the MATLAB code necessary to: 1. Add up the values of all of the elements. 2. Compute the sine of the given vector containing x-values (values may be assumed to be in radians). 3. Compute a running sum. A running sum is the summation of a sequence of numbers that is updated every time a new number is added to the sequence. So, the running sum of this x vector would be a new vector with the values $[1 \ 9 \ 12 \ 21 \ 21 \ 22]$.
3	Consider an M-by-N array of random numbers. Write the MATLAB code necessary to move systematically through the array, element by element, and set any value that is less than 0.2 to 0 and any value that is greater than (or equal to) 0.2 to 1.
4	Write the MATLAB code necessary to evaluate and plot the function $f(t) = 4 - t^2 e^{-3t}$ for the time range $0 \leq t \leq 3$ s. Use an appropriate step size for t to create a smooth curve. Label the axes on the plot appropriately.

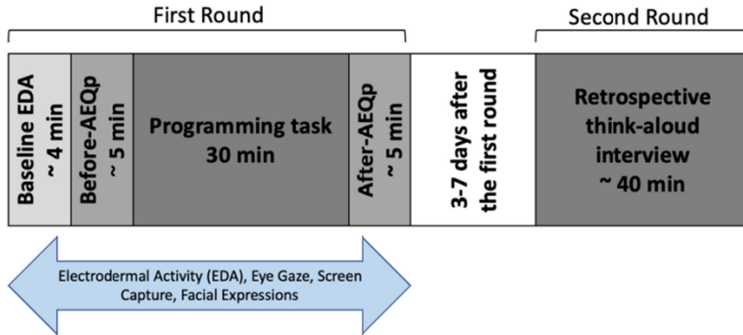


Fig. 1. Research design.

within an overarching design. For this study, the overall design involved multi-modal data collection. The data analysis was primarily qualitative followed by a triangulation phase. Figure 1 explains the research design specific to our research question.

4.4.1 Data Collection Methods. The following sections describe the three modes of data collection used for this study:

Achievement Emotions Questionnaire: Achievement Emotions Questionnaire (AEQ) is a questionnaire derived from the CVT [54] and is used extensively to understand students' emotions in the context of mathematics and language learning [18, 45]. Since we use CVT to guide different parts of our study, it is appropriate to use AEQ to collect self-report data in the current context. However, it must be noted that AEQ is not adapted for the purpose of learning programming. We adapted AEQ for the context of programming by rephrasing some words in the items (AEQp). For instance, "When I get the results of the test, my heart beats with pride" was rephrased to "When my program ran correctly, my heart beats with pride." Rephrasing items in a validated questionnaire requires rigorous re-validation using statistical techniques [76]. However, our study is qualitative with a small sample size. Hence, we did not validate the questionnaire

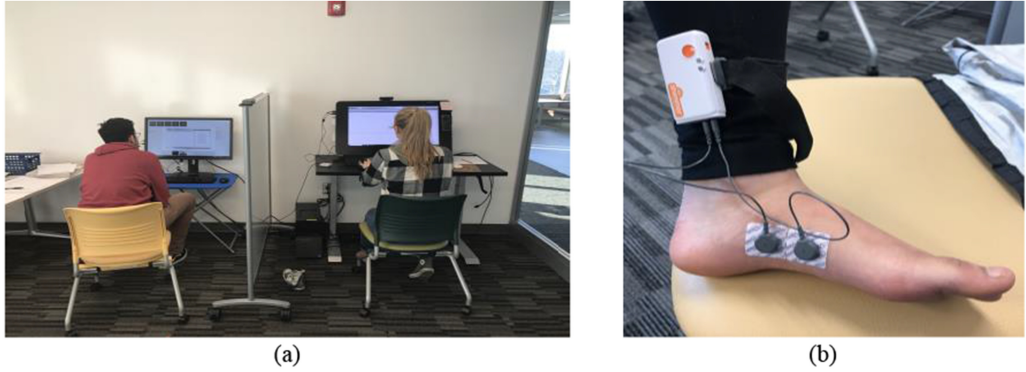


Fig. 2. (a) Experimental setup and (b) configuration of the Shimmer device.

using statistical techniques. Instead, we ensured the face validity of the adapted questionnaire by discussing it with peers, who had prior programming experience [49].

It is important to note that AEQp assesses emotions with respect to time. Emotions experienced before and at the start of the assessment or class are called prospective emotions, and emotions experienced after the assessment or class are called retrospective emotions [52]. For instance, anticipatory joy, hope, anxiety, and hopelessness are prospective emotions and outcome-related joy, pride, relief, and shame are retrospective emotions. Hence, some emotions in before-AEQp are not present in after-AEQp and vice versa. For instance, hope, hopelessness, and anxiety are in before-AEQp and not in after-AEQp, while relief is in after-AEQp and not in before-AEQp [51]. The items present in both before-AEQp and after-AEQp are described in Appendix B.

Participants' responses on before-AEQp and after-AEQp were used qualitatively to prompt participants' responses during the retrospective think-aloud interview. During the interview, participants were asked about their emotions before and after the programming task. The interview questions were based on situations that participants experienced during task [22] and their responses to the questionnaire. This data collection technique is called the participant-selection variant of the sequential explanatory mixed methods design during which results of the quantitative data inform the qualitative data collection [16].

Retrospective Think-Aloud Interview: Retrospective think-aloud interview is a verbalization technique in which participants verbalize their thoughts about a task that they had performed earlier [21]. The purpose of choosing the retrospective think-aloud technique was to avoid distraction during the task. Distraction could disrupt participants' cognitive process during the task and could subsequently induce emotions not caused by the task itself. One drawback of this technique is recall bias; that is, participants may not be able to accurately recall their experience of programming tasks. To minimize recall bias, we collected secondary data (eye gaze of participants on the screen, video of screen capture, and facial expressions) [17].

EDA: Collection of EDA requires non-invasive equipment (shimmer device) that has two electrodes that apply a constant low voltage to measure the electrical current between the electrodes (see Figure 2(b)). Electrodes are typically placed on either the fingers or the palms, because this configuration captures a pronounced EDA signal [14]. However, EDA is sensitive to noise, and since our participants had to type code during the programming task, we placed the electrodes on the participant's foot [20]. The placement of the electrodes is not the only prerequisite for the accurate collection of EDA. The participant should also be seated in a comfortable position with

Table 3. Steps for Inductive and Deductive Thematic Analysis

Steps	Description
1. Getting familiar with the data	• Transcribed the audio files. • Read and re-read the transcripts to make sense of these data
2. Development and reliability of the codebook	• Generated initial codebook using theory, literature, and data from pilot studies. • Established reliability with the help of another researcher
3. Assigning or generating initial codes	• Assign codes using codebook. • Generated a new code if it did not exist in the codebook and updated the codebook with new code.
4. Categorization	• Used theoretical framework and literature to categorize the code
5. Generating Themes	• Collated code into potential themes based on similar characteristics
6. Descriptive report and writing memos.	• Selected compelling excerpts. • Summarized and interpreted the excerpts using theory and literature. • Wrote memos during data analysis.

minimal movement (especially the limbs where electrodes are attached), should breathe normally without irregularities, and should avoid talking [14, 75].

4.4.2 Data Analysis Methods. The primary source of data was the retrospective think-aloud interviews, which were analyzed using thematic analysis. EDA data were analyzed using quantitative analysis procedures.

Thematic Analysis: The qualitative data collected through retrospective think-aloud interviews were analyzed using deductive and inductive thematic analysis. The deductive thematic analysis was driven by literature and the theoretical framework that guided the study [15], whereas in inductive thematic analysis the themes emerged from the data [49]. First, the researcher transcribed the interviews and read them multiple times to get familiar with these data [15]. Second, the initial code book for the overarching study was developed using NVivo, by using the two theoretical frameworks that guided the different stages of this study [57]. Specifically, for research questions addressed in this article, we used the idea of change in emotions to devise categories. That is, an emotion could either increase or decrease between the beginning and end of the programming task, or there was no change in emotion reported. Categories related to the reasons for change emerged inductively from these data. Third, the reliability of the codebook was established with the help of another researcher. Fourth, the codes were applied to these data. If the code for a certain theme was available in the code book, then it was used. Otherwise, a new code was generated and assigned to the transcript. Fifth, after the coding was complete, we started to categorize different codes into themes and sub-themes. We used the idea of change in emotions to devise categories. That is, an emotion could either increase or decrease between the beginning and end of the programming task, or there was no change in emotion reported. In this way, the themes emerged from the codes [15]. For a detailed description of steps for inductive and deductive thematic analysis see Table 3.

Quantitative Analysis: The EDA data consists of two components: tonic and phasic [14]. The phasic component of the EDA signal refers to a short, fast emotional, or cognitive response [14], and, hence, phasic EDA was used for the analysis. To account for interindividual differences in the EDA, range correction was calculated, which normalizes the phasic EDA between 0 and 1 [14]. To see how EDA data reflects the change in students' perception of their emotions before and after working on programming problems, we performed Wilcoxon signed rank test on range-corrected

EDA of baseline and after-Task. We chose Wilcoxon signed test, because the sample size was small, and we could not assume a normal distribution in the data [44].

4.4.3 Experimental Setup. To collect multi-modal data, we used iMotions [77], which is a software platform used to synchronize and integrate multiple types of data in one place (see Figure 2(a)). Multi-modal data were collected in two rounds, as discussed below.

First Round: During the first round of data collection, participants were asked to work for 30 minutes on the programming task. Before working on these problems, we asked participants to sit idle for 4 minutes to collect their baseline EDA using a non-invasive shimmer device (see Figure 2(b)). After 4 minutes, participants filled the before-AEQp. Once the participants completed the 30-minute programming session, they filled the after-AEQp. For more details on the steps of the first round see Figure 1.

Second Round: In the second round, a retrospective think-aloud interview was conducted with a video replay of the screen recording of the programming task. The interview was set up into two parts. In the first part, students elaborated on their emotional experiences of programming tasks. While in the second part of the interview, we showed participants their responses on before- and after-AEQp and asked some open-ended questions about why a certain emotion changed after completing the programming task. Specifically, the interview questions were as follows: (1) Your response shows a dip in [emotion] after completing the programming task. Why do you think that happened? and (2) I see that your [emotion] has gone up after completing the programming task. Could you explain why it has increased?

This study uses the responses of the second part of the retrospective think-aloud interview to answer the research question. For more complete details on retrospective think-aloud interview protocol see Reference [3].

We conducted the second round 3 to 7 days after the first round. This was necessary, because, in the retrospective think-aloud interview, we needed participants' responses on before-AEQp and after-AEQp to prompt the interview questions. Therefore, before conducting the interviews, we had to analyze each participant's responses on before-AEQp and after-AEQp to identify changes in their emotions between the beginning and the end of the programming task. Consequently, the second round could not immediately follow the first round on the same day.

5 FINDINGS

The primary data used to answer this research question were the retrospective think-aloud interview responses that were prompted by participants' scores on the before- and after-AEQp. As described above (Section 4.4.1) some emotions do not have items in both before-AEQp and after-AEQp (relief, anxiety, hope, and hopelessness), and hence, it was not possible to assess the qualitative change in these emotions. Moreover, participants did not elaborate on their experiences of anxiety during the retrospective think-aloud interview, and, hence, anxiety is not explained in the findings. The findings of this research question are summarized in Table 4.

5.1 Positive Activating Emotions

The following section elaborates on how participants' positive activating emotions (enjoyment and pride) changed because of working on the programming task.

5.1.1 Enjoyment. Findings from this study suggest that, for most participants, enjoyment decreased between the beginning and the end of the programming task. However, there is very little data on enjoyment increasing or on no change in enjoyment. Figure 3 and the table in

Table 4. Change in Emotions Before and After the Programming Task

Emotions	Change	Reason
Positive Activating Emotions		
Enjoyment	Decreased	- Faced challenges while programming
		- Could not transfer knowledge from class to the programming task
		- Had limited time
	Increased	- Found the problem on the programming task to be familiar
		- Outcome matched their expectations
Pride	Decreased	- Performance did not meet their expectations
	Increased	- Achieved more than what they expected
		- Successfully executed programs without help
Negative Activating Emotions		
Anger	Decreased	- Problems were easier than expected
	Increased	- Did not know what to do
		- Their performance did not meet their expectations
		- Repeatedly encountered errors
Shame	Increased	- Could not solve problems and attributed the failure to themselves
		- Should have known how to solve problems

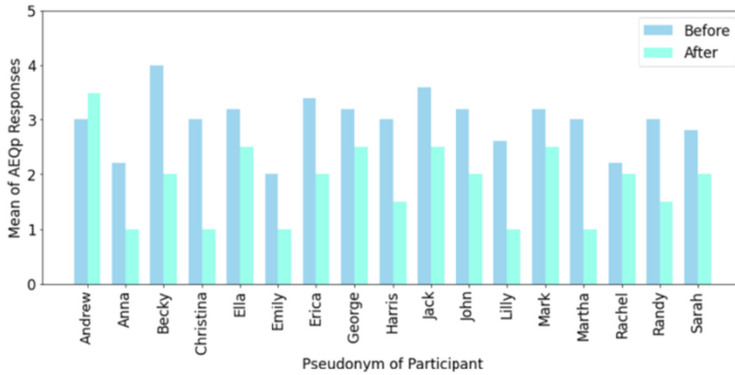


Fig. 3. Change in enjoyment before and after the programming task.

Appendix A show individual before- and after-AEQp scores. The difference in these scores was used to qualitatively infer whether the participant's perceived enjoyment increased or decreased between the beginning and the end of the programming task. These descriptive data are supplemented by participants' interview responses from the retrospective think-aloud interview.

Enjoyment Decreased: During the think-aloud interview, participants provided multiple reasons for decreased enjoyment.

(a) Faced Challenges While Programming

Some participants experienced a combination of dislike for programming, frustration, and the occurrence of challenges that led to decreased enjoyment between the beginning and the end of the task. For instance,

I don't like programming in general, but I think it went down because problem 1 gave me more errors than I anticipated ... It went down because I wasn't enjoying myself because of all the errors even though I knew it's one of those things where you know you can do it, but because there are so many things in the way, you get more and more frustrated, so you don't enjoy it. –Sarah

Like other participants in this sub-category, Sarah mentioned feeling frustrated in addition to decreased enjoyment between the beginning and the end of the programming task. Sarah did not like programming, but her self-concept was high when she mentioned that she knew she could do it. Despite having a high self-concept, she was not able to complete the task and experienced negative emotions [9, 17]. Furthermore, the control-value theory of achievement emotions states that failure instigates frustration [58]. Since Sarah encountered errors that led to the failure at the programming task, she experienced frustration combined with decreased enjoyment.

(b) Could Not Transfer Knowledge from Class to the Programming Task

This sub-group of participants failed to transfer the knowledge from one context (classroom) to another, in this case, a test-like programming task. For instance,

The biggest one for me is only really feeling confident about one of the four problems but then ending on a problem where like, I did not get anywhere. Do I find that fun, no. I think especially because I do learn things in class, but it's frustrating and discouraging when I cannot show that. And then I feel like if I can't have pride in my education, do I find that enjoying? No. –Christina

Christina explained that she had learned concepts in class, but she was unable to transfer that knowledge to another context, thus inducing negative emotions like frustration. Literature suggests that novices generally lack the ability to transfer knowledge from one context to another, and because of this they have difficulty completing academic assignments successfully [55]; hence, failure to perform successfully leads to frustration [58]. Furthermore, Christina mentioned that she was confident about solving one of the problems, indicating a high self-concept about that problem on the programming task. Christina's behavior aligns with the literature, which suggests that academic achievement influences students' self-concept [64]. However, when she did not perform adequately, she experienced a drop in her self-concept, and subsequently, her perceived enjoyment also decreased.

(c) They Had Limited Time

Another group of participants experienced lower enjoyment at the end of the task than at the beginning, because they thought there was not enough time to complete all four problems on the task. For example,

I think just the fact that I kind of went in with the thought that I'll probably finish these four problems. I'm just thinking that they were going to be easier. The first two were quite easy, the second two were pretty challenging, and I was a little frustrated with myself, so I don't think the task itself was that enjoyable –Harris

Harris had positive expectations about the task; that is, the problems were easy, and he would be able to finish all four of them. However, at the end of the task, he was frustrated that his performance did not meet his expectations, and hence he did not enjoy the task. Harris's experience is consistent with the control-value theory, which suggests that success is attained when students have sufficient control over the activity and its outcomes [58]. Since he encountered challenges

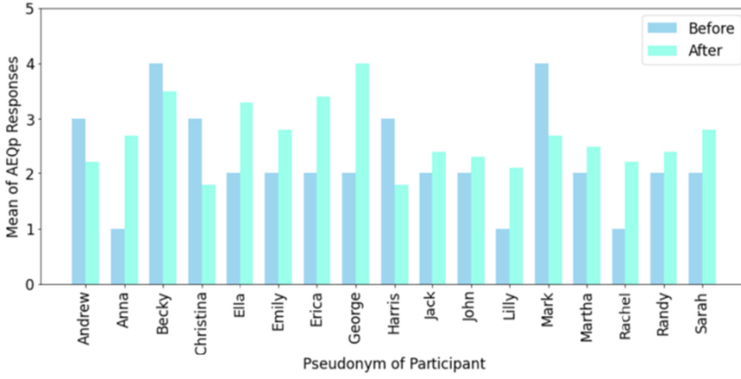


Fig. 4. Change in pride before and after the programming task.

while working on the task and found the problems to be hard, he felt frustrated and experienced decreased enjoyment between the beginning and end of the task.

Enjoyment Increased: In the retrospective think-aloud interview, some participants anticipated that they would enjoy the task, and they did report enjoyment when the task ended. For instance,

It could have been because it was more fresh in my mind, when I look the before one, the last problem that I had done was before spring break. And I felt like I did a pretty good job on it. So I enjoyed that one a little more than when I was working on this one. –Mark

Mark enjoyed working on one programming problem on the task, because he had previously done one of the four problems in class. Since the problem was familiar to him, he felt confident that he would be able to successfully complete that problem, and, hence, he felt in control of the situation. Mark's experience conforms with the control-value theory, which explains that people feel in control when they are able to successfully initiate or perform an activity [58]. However, Figure 3 shows that Mark's enjoyment decreased as he reported on the questionnaire. This contradiction between findings is a key aspect of multi-modal research, where divergence is also considered important because meaning making of different modes does not happen in isolation but together with each other [2].

Figure 3 shows an increase in enjoyment for Andrew. Andrew's retrospective think-aloud interview explains his increase in enjoyment before and after completing the task:

...I believe it came in knowing that I was actually done with the task, and I didn't have to do it anymore. So that was a thing with enjoyment. –Andrew

Overall, Andrew struggled with the task as shown in the following Section (5.3.1), hence, his increase in enjoyment after the task is associated with being done with the task, instead of the problems on the programming task.

5.1.2 Pride. During the retrospective think-aloud interview, participants reported that their pride in themselves after finishing the programming task either decreased or increased.

Figure 4 and the table in Appendix A show individual before- and after-AEQp scores. The difference in these scores was used to qualitatively infer whether the participant's perceived pride increased or decreased between the beginning and the end of the programming task. These

descriptive data are supplemented by participants' interview responses from the retrospective think-aloud interview.

Pride Decreased: Positive activating emotions like pride relate positively with the effort students expend in an academic task [51]. Hence, students feel proud if the outcomes of their efforts meet their expectations and vice versa. Some participants in this study experienced a decline in their perceived pride when their performance did not meet their expectations. For example,

How can I feel good about myself if I literally just failed, especially the red messages and stuff, the error messages, like no one likes getting an error but it does really hurt your pride, because what you thought was going to work didn't so now it's up to you to try to figure out how to fix it but if you don't necessarily know how to do that. –Christina

Christina thought the code she had written would work, but she encountered errors that she could not fix. When she realized that she did not know how to fix the errors, her pride decreased. Participants also experienced decreased pride, when they thought they had not met the standards they had set for themselves. For instance, Harris explained how his pride decreased because he had not met his own standards as follows:

Just because I think I hold myself to fairly high standards and so when I don't kind of reach those standards, I get pretty upset with myself. Yeah! So I think again, like, I was pretty frustrated with myself. –Harris

He thought he had enough time to finish all the problems on the programming task, but his inability to finish the problems made him feel frustrated. Harris's experience can be explained by the social cognitive theory, which explains that people react negatively when they fall short of or violate personal standards [5]. Both Christina and Harris had a high self-concept about programming at the beginning of the programming task ("... what you thought was going to work ..." and "... I hold myself to fairly high standards ...," respectively). However, their self-concept changed because they both encountered challenges during the programming task, hence, decreasing perceived pride. The change in the self-concept for both Christina and Harris shows the reciprocal relation between academic achievement and self-concept, as mentioned in References [19, 29, 30, 64].

Pride Increased: The overarching theme from these data is that participants felt proud of their performance if they met or exceeded the expectations they had from the task. Data from the study uncovered different ways in which participants met or exceeded their expectations, thus leading to increased pride. Moreover, it is important to mention that every participant in the study had a different conception of what success meant to them.

(a) *Achieved More Than What They Expected*

Anna explained that her pride increased because she had achieved more than she had expected before the task started.

I was pretty proud of myself, I knew I had done more than I thought I was going to do, and especially because I did finish the one, completely [...] I find a lot of times I get stuck is with the first one where we all have one error somewhere and I can't find that one error and that is the most frustrating thing in the world. –Anna

Generally, when Anna encounters errors on a programming problem, she gets frustrated trying to fix the errors, and then she cannot complete other problems that she is supposed to work on. During the programming task, Anna was able to avoid this hindering behavior, and, hence, she

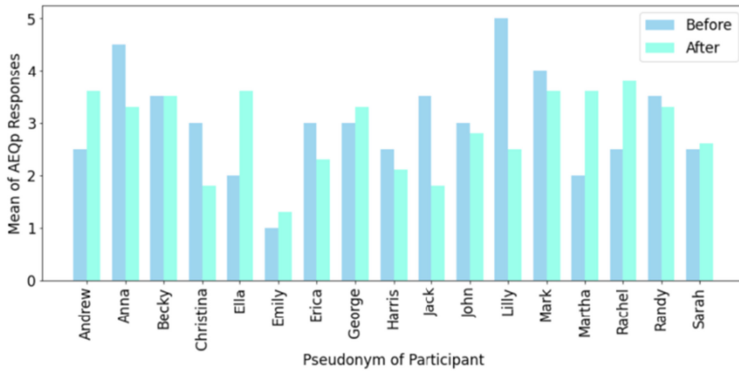


Fig. 5. Change in anger before and after the programming task.

achieved more than her expectations, leading to increased pride. Anna's experience conforms with the control-value theory, which suggests that pride is induced when success is perceived due to internal control by the student [51, 58]. In this case, Anna evaluated the factors that could hamper her success, overcame those factors, and achieved something, leading to a sense of increased pride [48, 58].

(b) Successfully Executed Programs Without Help

In ENGR 132, students mostly work in pairs on their programming assignments. They also have help readily available (for instance, access to teaching assistants). Sarah explained that this programming task gave her the opportunity to test her programming skills in an environment that did not provide any help in the form of peers and teaching assistants. In such an environment, she was able to fix syntax errors and successfully execute programs, and that resulted in increased pride.

I don't know if it was pride but it made me feel kind of better about myself 'cause normally my programs don't run as I want them to and then normally I ask my peers but in this case I had to fix it all by myself. So after I fixed it all and it ran, it made me feel better because I was, like, I finished this all by myself. So this is something I can do, which was not something I tend to feel when I'm done programming, like tasks and problem sets. –Sarah

Since Sarah was able to fix errors by herself, that is, she felt in control of her circumstances, and, hence, she experienced increased pride between the beginning and end of the programming task. Sarah's experience conforms with the control-value theory, which suggests that students experience positive emotions when they feel in control of their circumstances [51]. Moreover, Sarah had a low self-concept going into the task, but her achievement on the task increased her self-concept and subsequently her pride increased at the end of the task [64].

5.2 Negative Activating Emotions

The following sections describe how participants' anger and shame changed between the beginning and end of the programming task.

5.2.1 Anger. During the retrospective think-aloud interview, participants explained why their anger increased or decreased between the beginning and the end of the programming task.

Figure 5 and the table in Appendix A show the mean before- and after-AEQp scores for individual students. The difference in these scores was used to qualitatively infer whether the participant's perceived anger increased or decreased between the beginning and the end of the programming

task. These descriptive data are supplemented by participants' interview responses from the retrospective think-aloud interview.

Anger Increased: Before the programming task, most participants anticipated that they would not feel angry. However, their perceived anger increased between the beginning and end of the task. These participants encountered unanticipated obstacles during the task, which frustrated them, subsequently leading to increased anger. According to the control-value theory, frustration and anger are likely to occur together [58]. Anger and frustration are aroused when working on a task, requiring so much effort that the resulting experience becomes detrimental to the student [58].

(a) Repeatedly Encountered Errors

The following excerpts by Sarah show why she experienced increased anger.

Well, I was annoyed, I was frustrated. As I said before, it was such a simple task but I couldn't finish it in the way I wanted to, or in the speed I wanted to. In like every error that kept on coming up I was like Again? Or like Come On! I thought this was as simple as I made it out to be but it's not. –Sarah

Sarah explained that she was expecting the task to be simple, but it was not. Moreover, she kept on encountering errors while coding. Hence, she did not feel in control of her actions (coding), and outcomes (could not finish the task due to errors), and hence experienced frustration, leading to increased anger after the task was finished.

(b) Not Knowing What to Do

The following excerpt by Ella explains her emotions when she realized she did not know what to do.

I think personally not knowing how to do something, it's very frustrating to me. So coming across that question where I had no idea how to even start it, I was probably pretty upset about that. I just wasn't happy about the last question but I think that's why I saved it for the end is because I knew like I didn't want to. –Ella

Ella, however, felt out of control for a different reason. She experienced increased anger at the end of the task and frustration while working on the programming task because she was unable to finish the last problem on the task. Ella's experience could also be explained by the peak and end rule, which posits that a person judges their overall experience based on how they felt at the peak of their experience and at the end of the experience [32].

(c) Their Performance Does Not Meet Their Expectations

Some participants had set certain standards for their performance, when they were unable to meet those standards, they experienced increased anger after the task ended. Anna mentioned that she sets high standards for herself, and when she does not match those standards, she gets angry.

I was upset that my program had errors but it's always more anger with myself I think than it is at anyone else. I'm a pretty self-motivated learner and I usually feel like the standard I perform at is my responsibility so that there's no excuse if, I can sit here and tell you the way they teach doesn't match with my learning style but in the end, that means that I should be going out and searching for, I should be finding a way to teach it to myself that matches up with my learning style because I do realize that you have a room full of a hundred people, you can't teach it to fit everyone's style. –Anna

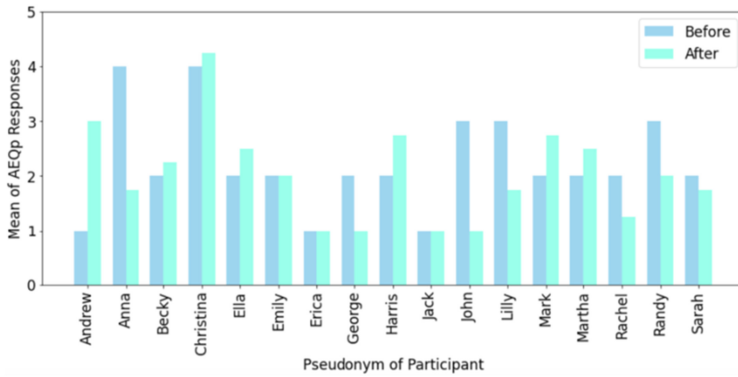


Fig. 6. Change in shame before and after the programming task.

Anna's experience could be explained by the social cognitive theory, which explains that people experience negative emotions when they fall short of their self-set personal standards [5]. She also added that as an intrinsically motivated learner, she is responsible for her own learning. Hence, she does not like to blame external factors for her lack of learning. She would rather search for material and teach herself. Anna's attainment of her goal or self-set standard was interrupted, and because she does not attribute her learning to external factors (including the class and teachers), she feels angry at herself for not reaching her standards. Anna's example contradicts the attribution theory, which states that when we fail or make mistakes, we attribute the cause to external factors instead of blaming ourselves [70]. Although the retrospective think-aloud interview shows that Anna's anger increased, Figure 3 shows that her average anger score decreased before and after the task, hence, showing divergence in these data.

Anger Decreased: Some participants were expecting the programming task to be hard, and hence their expectation of anger was high. However, at the end of the task, they realized that their anger at the end of the task was less than what they have expected to experience before the task started, as shown by an excerpt from Lilly.

I was expecting the task to be a lot harder. Okay. Because like some of them I found easier and some of them were harder, so I was like really annoyed at that. But then afterward it was a lot better. –Lilly

Before starting the programming task, Lilly was expecting the programming problems to be hard. She did find some problems to be hard, but some problems were also easy. Overall, she explains that the problems were easier than she expected and hence, her anger decreased coming out of the task.

5.2.2 Shame. During the retrospective think-aloud interview, participants explained why their shame increased or why there was no change in shame before and after the programming task.

Figure 6 and the table in Appendix A show the mean before- and after-AEQp scores for individual students. The difference in these scores was used to qualitatively infer whether the participant's perceived shame increased or decreased between the beginning and the end of the programming task. These descriptive data are supplemented by participants' interview responses from the retrospective think-aloud interview.

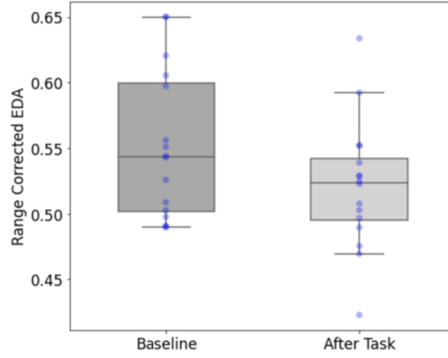


Fig. 7. Baseline vs. after-Task range-corrected EDA.

Shame Increased: Most participants experienced shame because they experienced failure and attributed the failure to themselves instead of external causes, as shown in the excerpts by Randy and Christina.

Oh, absolutely because I felt like I should have known the majority of this and I flat out didn't so yeah, it was very embarrassing to myself [...] I mean I'm being recorded, I'm being tested and so, performing the way I did, made me feel pretty bad. –Randy

I just look dumb, I couldn't figure some of them out especially I think the plotting one, its literally plotting, and I looked up the help function, I still couldn't get it so that was embarrassing. I don't like when things don't work, I mean, because all thing has been covered. I should know how to do it but I couldn't figure it out so yeah, I felt that's embarrassing. –Christina

Randy said, "I should have known the majority of this"; Christina said, "I should know how to do it." These experiences of Randy and Christina conform with the control-value theory, which suggests that shame is induced when failure is judged to be caused by oneself [50]. Both attributed the cause of failure to themselves. However, Christina explicitly called herself stupid, because she was unable to successfully solve the problem despite using help. According to participants' responses on the before- and after-AEQp, Christina's shame did increase slightly. However, Randy's shame decreased showing divergence between the two data sources.

5.3 Electrodermal Activity Data and Emotions

For the purpose of analysis, EDA data from 16 participants were used, because one participant's EDA data was not captured by the shimmer device. For each participant, the EDA data is divided into different sections, depending on the part they engaged with (Baseline, before-Task, During-Task, and after-Task) (see Figure 1). Literature suggests that participants' EDA may decrease after they have worked on a cognitively challenging task [67]. Hence, we performed the Wilcoxon signed rank test between the range-corrected EDA of baseline and after-Task. However, the results suggest a non-significant difference between the two variables ($Z = 33.5$, $p = 0.13$), as shown in Figure 7.

Participants' EDA data could be explained by considering the progression of parts as shown in Table 5 and by individual participants' data. Even though the result of the Wilcoxon signed rank test is not statistically significant, we find value in looking at the EDA graphs from individual students, to show how EDA is affected by students' actions and behaviors during the task. Different

Table 5. Mean and Standard Deviation of Range-corrected Phasic EDA

Pseudonym	Baseline		before-AEQp		MATLAB Session		after-AEQp	
	M	SD	M	SD	M	SD	M	SD
Andrew	0.509	0.073	0.548	0.096	0.391	0.046	0.552	0.131
Anna	0.605	0.14	0.547	0.191	0.49	0.15	0.423	0.195
Becky	0.65	0.068	0.484	0.063	0.484	0.063	0.552	0.128
Christina	0.526	0.095	0.562	0.144	0.455	0.093	0.529	0.139
Ella	0.544	0.08	0.53	0.101	0.528	0.088	0.592	0.123
Emily	0.597	0.096	0.529	0.144	0.529	0.144	0.523	0.125
Erica	0.65	0.062	0.537	0.133	0.549	0.037	0.529	0.113
George	0.551	0.051	0.502	0.147	0.542	0.082	0.47	0.122
Harris	0.62	0.077	0.381	0.161	0.595	0.06	0.633	0.126
Jack	0.491	0.137	0.458	0.171	0.576	0.102	0.525	0.153
John	0.503	0.125	0.388	0.174	0.533	0.138	0.476	0.141
Lilly	0.498	0.096	0.476	0.166	0.52	0.149	0.539	0.213
Martha	0.556	0.108	0.489	0.16	0.455	0.123	0.508	0.142
Rachel	0.491	0.597	0.504	0.138	0.585	0.112	0.503	0.129
Randy	0.543	0.077	0.548	0.112	0.499	0.086	0.497	0.149
Sarah	0.49	0.051	0.584	0.119	0.514	0.092	0.49	0.139

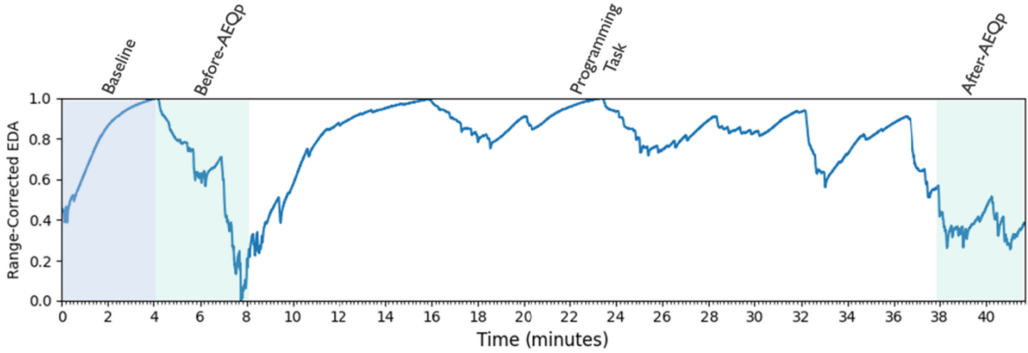


Fig. 8. Andrew's range-corrected EDA over time.

participants handle cognitive and emotional stimuli differently, as evidenced by the EDA data shown in Table 5 and as explained in the following two exemplars.

5.3.1 Andrew. Compared to the baseline ($M = 0.509$, $SD = 0.073$), Andrew's mean range-corrected phasic EDA for the before-Task period was $M = 0.584$, $SD = 0.096$. Since the mean values were so close, it was inferred that Andrew experienced no arousal between the two periods. However, during the programming task, there was a substantial decrease in Andrew's arousal response ($M = 0.391$, $SD = 0.046$), indicating most likely physiological arousal due to performing cognitively challenging tasks as Andrew was engaged in high-level algorithmic thinking. Change in Andrew's EDA over time can be seen in Figure 8.

Literature suggests that programming involves higher cognitive demands [59]; hence, Andrew experienced low arousal and high cognition during the programming task. Immediately after the programming task, Andrew's arousal levels were substantially higher ($M = 0.522$). Findings from

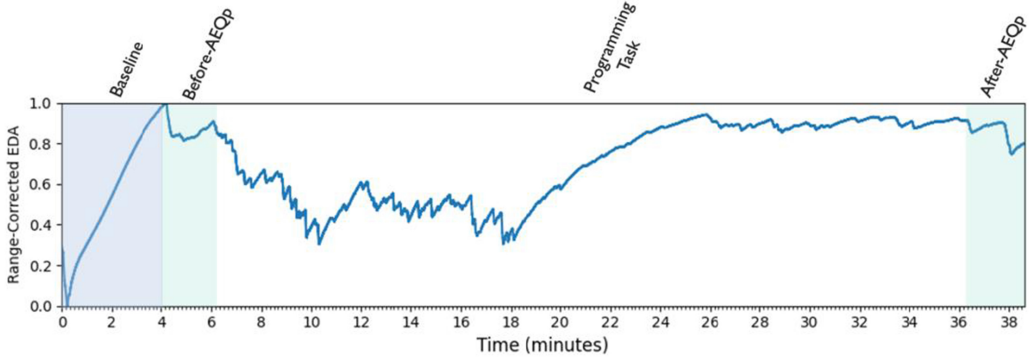


Fig. 9. Becky's range-corrected EDA over time.

Andrew suggest an increased sympathetic activity (fight and flight response) during the application of a mental load compared to the baseline [56, 63].

Andrew's responses on the before- and after-AEQp (Figure 3, 4, 5, and 6) show a decline in positive activating emotions (enjoyment and pride), and an increase in negative activating emotions (anger and shame). These data align with his high arousal levels shown by his EDA. These data also align with Andrew's response on the retrospective think-aloud interview shows the challenges he faced while working on the programming task:

And I'm back over the back, over the problem set to make sure that I'm not going to part without hovering everything that was asked of me. [...] I'm right here. I'm right here. I notice I got to a part that didn't know, didn't understand completely [...] Frustration because it didn't know how to approach it at first [...] Yes. Okay. And then, um, and then, um, and then after a google, did I notice that, um, that is a command in, you know, MATLAB that actually does it. So after that I was content.

Andrew's response on the retrospective think-aloud interview confirms these findings as Andrew struggled while working on these problems. He went back and forth between writing code and looking at help resources many times to solve the problems on the task.

5.3.2 Becky. Becky's baseline values were much higher than any of the other events that followed ($M = 0.650$, $SD = 0.068$). This difference suggests that Becky was undergoing emotional or cognitive stressors before the start of the experiment. Similar work by Reference [67] indicates the importance of anticipatory responses in electrodermal activity data. Changes in Becky's EDA over time can be seen in Figure 9.

Becky's responses on the before- and after-AEQp shows a decline in positive activating emotions (enjoyment and pride), no change in anger, and an increase in shame. Becky's response to the retrospective think-aloud interview suggests that she executed code multiple times without making any changes to the code, pointing to the possibility that she may have been emotionally distressed before starting the programming task. The following excerpt from Becky during the retrospective think-aloud interview points to this behavior:

And at this point I made sure to check if it was correct. So still just checking it. I like to check things multiple times so that way I know that somehow it was not just a fluke and that I know it was giving me the answers that I want so I think. Now I am just checking it over to make sure that I did everything and now I am moving on to the second one.

6 DISCUSSION

The key finding from this research question is that students may experience increases in some emotions and decreases in other emotions in response to the same events. When participants experienced decreased enjoyment between the beginning and the end of the task, they also experienced increased anger most of the time. Participants experience decreased enjoyment and increased anger, because they encountered numerous challenges during the programming task. Similarly, when participants experienced decreased pride between the beginning and the end of the programming task, they also experienced increased shame, mainly because they fell short of the expectations or standards, they had set for themselves. These findings align with literature that suggests that there is a consistent negative correlation between anger and enjoyment, and between pride and shame in academic settings [51]. However, existing correlational research fails to describe the thick rich descriptions participants provided in this study for the increase and decrease in emotions between the beginning and the end of the task.

Many participants mentioned feeling frustrated in addition to decreased enjoyment between the beginning and the end of the task. Since this group of participants failed at achieving what they set out to do, they experienced frustration. Participants' frustration during the programming task is consistent with the literature [10, 37]. However, this study also explained how participants' enjoyment decreased because of the frustration they experienced.

Negative activating emotions (anger and shame) have a mixed effect on participants. On the one hand, these emotions could seriously undermine intrinsic motivation, thus hampering learning and performance [51]. On the other hand, the activating nature of these emotions could encourage students to deal with the events that caused the emotions, thus strengthening extrinsic motivation [51, 52]. Research suggests that negative emotions may be most beneficial for tasks that require careful attention to detail, to prevent or detect errors [51]. Hence, negative activating emotions may not necessarily impair student learning and performance.

6.1 Expectations of Success and Emotions

Most participants in this study had positive expectations and high self-efficacy beliefs before starting the task. When their expectations did not yield positive outcomes, their enjoyment decreased between the beginning and the end of the task. These findings align with previous literature that suggests that negative self-efficacy beliefs yield negative outcomes [36, 37]. Findings from this study also suggest that there were some exceptions, where participants' positive expectations yielded positive outcomes, but they still experienced negative emotions and decreased enjoyment during the task. These participants struggled significantly during the programming task and engaged in repeated trial-and-error before finally succeeding on a programming problem. Moreover, the findings reflect the reciprocal relation between self-concept and educational experience (academic achievement in the current context) and how their interaction affects emotions. Participants who had high self-concepts at the start of the task, experienced negative emotions when they encountered failure and ended up with low self-concept. Similarly, participants with low self-concepts experienced positive emotions when they performed beyond their expectations, thus increasing their academic self-concept. Pride positively relates to the effort students expend and the self-regulation of their learning [51]. Pride is control dependent, that is, it is induced by self-related factors that influence outcomes, for instance, ability or lack thereof [58]. Students feel proud when they succeed in academic tasks, thus leading to enhanced motivation [48].

6.2 Experiencing the Same Event Differently

Findings from this study also suggest that different participants can experience the same event differently. For instance, one participant may experience increased pride between the beginning

and the end of the task, because they were able to complete most of the problems on the task. However, another participant experienced decreased pride, because they were unable to complete all the problems on the programming task. Literature suggests that similar experiences could be interpreted differently, with different emotional reactions [6].

6.3 Experiencing Multiple Emotions at the Same Time

Some participants experienced negative activating emotions like anger and shame, because they attributed the cause to themselves. Participants experienced anger and shame when they had set certain standards for themselves or had positive expectations about their ability to successfully complete the task. These findings align with literature that suggests that people experience anger when they are prevented from reaching their goals [7]. Moreover, anger and frustration normally occur together. For instance, participants reported feeling frustrated in addition to increased anger during the task. It is likely that participants experienced frustration during the programming task, which led to increased anger. These findings agree with CVT, which suggests that activities and outcomes are linked, and are influenced by emotions and the antecedents of emotions [58].

6.4 Convergence and Divergence in Multi-modal Data

For the most part, we see that the different data sources in our findings converge and align with each other. However, there are a few instances in our findings where we see divergence. For instance, Figure 3 shows that Mark's enjoyment decreased as he reported on the questionnaire. Similarly, Anna stated that her anger increased as a result of working on the programming task. However, Figure 5 shows that her anger before and after the task decreased. This contradiction between findings is a key aspect of multi-modal research, where divergence is also considered an important finding [2]. There could be multiple explanations for this divergence. First, both types of data (AEQp and retrospective think-aloud interview) were collected on different days and hence, their before- and after-AEQp responses differ from their interview response (recall bias). Second, students are generally not able to articulate their emotions properly [27] and hence the divergence in these data. There could be more explanations that the current sources of data are not able to reveal. Adding an additional source of data (for example, eye gaze), may provide additional reasons for observing divergence in our findings.

6.5 Electrodermal Activity and Emotions

For this study, we also analyzed participants' EDA data before and after completing the programming task. Overall, findings suggest that there were no significant differences in participants' range-corrected EDA values between the baseline and the after-AEQp periods. However, the EDA data for two participant exemplars explained in the findings point to the fact that there are individual differences in how students physiologically respond to a cognitively challenging task like programming. This aligns with the literature that suggests that the EDA data helps in identifying emotional arousal due to programming [27]. For triangulation purposes, we corroborated EDA data with participants' responses from the retrospective think-aloud interview for two exemplars. For both exemplars, participants' responses on the interview, and before- and after-AEQp responses aligned with the quantitative EDA findings. However, more in-depth and fine-grained analysis along with another type of data source may help provide a richer description of students near real-time emotional, motivational, and cognitive behaviors during a programming task [14].

7 LIMITATIONS AND FUTURE DIRECTIONS

This study has a few limitations discussed in this section. First, EDA data could be affected by caffeine usage and medicine intake [14]. However, these limitations were not considered, because the average range-corrected EDA for the baseline was higher than the average range-corrected EDA

for before-Task period. Second, a detailed event-based analysis of EDA could not be done, because participants worked on programming problems in random order according to their choice. By looking closely at two exemplars (Becky and Andrew), we found in-depth individual differences. This means that by using a multi-modal approach [2] more thorough and in-depth description of students' emotional behaviors could be obtained. Our team is in the process of developing a rigorous triangulation protocol involving multi-modal data (e.g., self-report, clickstream, and biometric) [71]. For more in-depth analysis, we aim to qualitatively analyze multiple sources of data along the temporal dimension (electrodermal activity, eye gaze, clickstream, and qualitative excerpts from the retrospective think-aloud interview). These data would provide multiple perspectives on events happening at a certain point in time. At this stage, a rigorous quantitative data analysis was not suitable because of small sample size. However, we performed Wilcoxon signed rank test on range-corrected EDA of baseline and after-Task, which did not show a significant change in emotions.

Third, there are some other confounding factors for EDA data collection, which need to be accounted for while collecting EDA. For instance, EDA may be affected by medical history of heart conditions, caffeine, drug consumption, and traumatic brain injury to name a few [14]. However, these prerequisites were not considered for two reasons: to keep the study design simple and to prevent the sample from reducing further. That is, using these parameters as exclusion criteria would have likely reduced the sample size of the study further.

Fourth, a unique aspect of this study is that participants' responses to the AEQp were used as prompts for a semi-structured qualitative interview. Since this study was primarily qualitative, and due to the small sample size, we did not conduct statistical analyses using the AEQp data. Moreover, the AEQp is also not validated for programming. Our team is currently in the process of collecting data for rigorously validating the AEQp for programming using statistical methods [4]. Once the questionnaire is validated, a large-scale study could be conducted to understand student emotions in the context of programming.

Fifth, the retrospective think-aloud interviews were conducted three to seven days after the programming session. Due to this, participants might have faced difficulty to accurately recall their experience of programming tasks while interviewing. However, to help them recall and minimize the bias, we replayed the video of their programming task annotated with their eye gaze and facial expressions during the interview.

8 IMPLICATIONS FOR CLASSROOM PRACTICE

In this study, a few participants experienced changes in emotion due to their lack of problem-solving skills. For instance, Ella mentioned that she could not figure out how to start a task (Section 5.2.1). To help students learn problem-solving skills for programming, instructors may use competence training to teach students programming [50]. One such problem-solving strategy is the “**paper-and-pencil programming strategy**” (PPS). PPS is a way to represent the logic of a computer program by using any representation that can be created on paper [35]. The representation style for the program in PPS is not rigid, it could be textual or simple diagrams [35]. One representation style that aids PPS could be used stand-alone to convert program description into visualization is the “flowchart” [35]. Flowcharts represent programming concepts through icons and connections among icons show the flow of program logic. Kim et al. used PPS to teach computational thinking to non-computing major students and found that PPS considerably helped students understand computational thinking concepts and elevated their interest in computer science [35]. Nowadays, students use devices like iPad and tablets pc to take notes in class. PPS could easily be transferred in the current classroom environment, because these devices support the drawing of flowchart-like structures or any other representation for creating the program logic.

Another important finding that suggests the lack of problem-solving skills among participants was their difficulty in transferring classroom knowledge to solve programming tasks. As Christina mentioned, it was frustrating and discouraging for her to not be able to show the learnings of class in solving the tasks (Section 5.1.1). A strategy that may help students to transfer learning to solve real programming problems and enhance their involvement during class is “problem-based learning” [43]. This strategy is widely used to master a concept/topic, which could also be applied in teaching programming. In programming, instructors may choose a set of programming problems for each concept/topic and after teaching that topic, the instructor may ask students to solve those problems at the end of class. In problem-based learning, instructors can also play a role by providing “Scaffolding.” Through scaffolding, instructors can provide hints, feedback, or answer students’ queries to help them learn and master the skills [72].

Students’ expectation of success and failure may influence their emotions in academic settings, as shown by the findings of this study. For instance, Initially, Harris had positive expectations about the task; however, at the end of the task, he was frustrated that his performance did not meet his expectations (Section 5.1.2). Different students conceptualize success differently. Moreover, students’ expectation of success is also shaped by societal norms and their interaction with family and peers [50]. Instructors may need to implement competency-based learning that shifts the focus from assessing the overall success of class to assessing individual learning [69]. By applying the competency-based learning in computer programming classes, instructors can redefine success to be more inclusive of students with different competencies and refrain from comparing their performance. However, it is important to acknowledge that implementing competency-based learning is a hard undertaking, because it requires not only a shift from a teacher-centered to learner-centered paradigm but also resources for implementation (for example, teacher training and preparation) [69].

9 CONCLUSION

This study contributes to the body of knowledge in two main ways. First, the findings provide a fine-grained and nuanced understanding of the reasons why students’ emotions change as a result of working on a programming task. Second, we adopted a novel multi-modal methodology that uses multiple modalities. For this study we used two self-report methods (retrospective think-aloud interview and AEQp), and one biometric source of data (electrodermal activity). The key finding of this qualitative study suggests that students may experience increases in some emotions and decreases in other emotions in response to the same events. This change in emotions is because of certain standards students set for themselves, or they had expectations about their ability to successfully complete the task. Whenever, their expectations did not yield positive outcomes, or they perform beyond their expectations, their emotions change between the beginning and the end of the task. These findings agree with the control-value theory, which suggests that activities and outcomes are linked, and are influenced by emotions and the antecedents of emotions [58]. Analysis of EDA data suggests a non-significant difference between the beginning and the end of the programming task. Upon closer examination of these data, differences were found during the programming task as diverse students handled cognitive and emotional stimuli differently.

APPENDICES

A APPENDIX

before- and after-AEQp Responses for Each Participant

Pseudonym	Positive Activating Emotions				Negative Activating Emotions			
	Enjoyment		Pride		Anger		Shame	
	Before	After	Before	After	Before	After	Before	After
Andrew	3.0	3.5	3.0	2.2	2.5	3.6	1.0	3.0
Anna	2.2	1.0	1.0	2.7	4.5	3.3	4.0	1.75
Becky	4.0	2.0	4.0	3.5	3.5	3.5	2.0	2.25
Christina	3.0	1.0	3.0	1.8	3.0	1.8	4.0	4.25
Ella	3.2	2.5	2.0	3.3	2.0	3.6	2.0	2.5
Emily	2.0	1.0	2.0	2.8	1.0	1.3	2.0	2.0
Erica	3.4	2.0	2.0	3.4	3.0	2.3	1.0	1.0
George	3.2	2.5	2.0	4.0	3.0	3.3	2.0	1.0
Harris	3.0	1.5	3.0	1.8	2.5	2.1	2.0	2.75
Jack	3.6	2.5	2.0	2.4	3.5	1.8	1.0	1.0
John	3.2	2.0	2.0	2.3	3.0	2.8	3.0	1.0
Lilly	2.6	1.0	1.0	2.1	5.0	2.5	3.0	1.75
Mark	3.2	2.5	4.0	2.7	4.0	3.6	2.0	2.75
Martha	3.0	1.0	2.0	2.5	2.0	3.6	2.0	2.5
Rachel	2.2	2.0	1.0	2.2	2.5	3.8	2.0	1.25
Randy	3.0	1.5	2.0	2.4	3.5	3.3	3.0	2.0
Sarah	2.8	2.0	2.0	2.8	2.5	2.6	2.0	1.75

B APPENDIX

Achievement Emotions Questionnaire for Programming

before-AEQp

The following questions pertain to feelings you may experience BEFORE starting a programming task. Please indicate how you are feeling, before starting the programming task.

1– Strong Disagree to 5 – Strongly Agree

		1	2	3	4	5
Task Enjoyment						
1.	I am looking forward to the programming task					
2.	I am looking forward to demonstrating my knowledge					
3.	Because I enjoy preparing for the programming task, I’m motivated to do more than is necessary.					
4.	Because I look forward to being successful, I have studied hard					
5.	Before starting a programming task, I sense a feeling of eagerness					
Task Pride						
6.	I am so proud of my preparation that I want to start the programming task now					
Task Anger						
7.	I get angry over time pressures which don’t leave enough time to prepare.					
8.	I get angry about the amount of material I need to know					
Task Shame						
9.	I can’t even think about how embarrassing it would be to fail the programming task.					

after-AEQp

The following questions pertain to feelings you may experience AFTER completing the programming task. Please indicate how you feel, typically after completing a programming task.

1 – Strong Disagree to 5 – Strongly Agree

		1	2	3	4	5
Task Enjoyment						
1.	My heart is beating faster with joy					
2.	I am glowing all over					
Task Pride						
3.	I am very satisfied with myself					
4.	I am proud of myself					
5.	Thinking about my success is making me feel proud					
6.	I am proud at how well I mastered the programming task					
7.	When my program ran correctly, my heart beats with pride					
8.	After the programming task I am feeling ten feet taller because I'm so proud					
9.	I am walking out of the programming task with the look of a winner on my face					
Task Anger						
10.	I am fairly annoyed.					
11.	I am angry because my program had errors					
12.	I wish I could yell at the compiler/interpreter					
13.	I wish I could freely express my anger					
14.	My anger is making the blood rush to my head					
15.	I am feeling so angry, I am feeling hot and flushed					
Task Shame						
16.	I am feeling ashamed					
17.	My performance on programming tasks is embarrassing me					
18.	When my program fails, I would prefer not to face my teacher again					
19.	When others find out about my poor programming ability, I start to blush					

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Received 28 February 2023; revised 25 December 2023; accepted 4 January 2024